

Chapter 1

New Beginnings

Definition 1.0.1: Metric Space

A metric space (X, d) is a set X with a function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ such that

1. For $p, q \in X$, $d(p, q) = 0 \iff p = q$;
2. $d(p, q) = d(q, p)$; and
3. the triangle inequality— $\forall p, q, r \in X$, $d(p, r) \leq d(p, q) + d(q, r)$ —holds.

Example 1.0.1

1. $X = \mathbb{R}$ so $d(a, b) = |b - a| = |a - b|$;
2. $X = \mathbb{R}^n$ with d as Euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (y_i - x_i)^2}$;
3. on $\mathbb{R}^n$ we have $d_1(x, y) = \sum_{i=1}^n |y_i - x_i|$ and $d_\infty(x, y) = \max(|y_i - x_i|)$.

Definition 1.0.2: Open

$S \subset X$ is open if $\forall p \in S, \exists r > 0$ s.t. $B_r(p) \subset S$.

Visually we can represent this with the following:

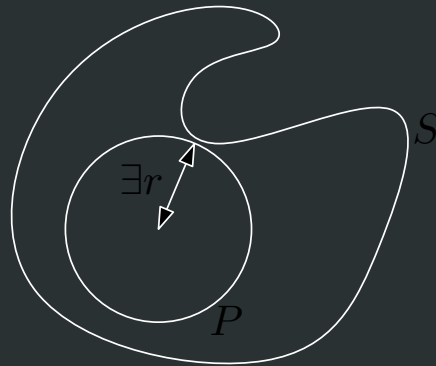


Figure 1.1: open

Definition 1.0.3: Open ball

Given a metric space (X, d) , $p \in X$, $r > 0$, the open ball (or neighborhood) of radius r around p is $B_r(p) = \{q \in X \mid d(p, q) < r\}$.